79 distinct compositions). In contrast, there are only two materials in the labeled fine-tuning dataset with such high bulk modulus values. Note that both MatterGen and screening produce multiple structures per composition that are unique according to our definition (Appendix D.3.1) but could potentially be alloys or solid solutions [57].

2.6 Designing low-supply-chain-risk magnets

Most materials design problems require finding structures satisfying multiple property constraints. MatterGen can be fine-tuned to generate materials given any combination of constraints. Here, we showcase its ability to tackle materials design problems with multiple constraints by searching for low-supply-chain-risk magnets. Since many existing high-performing permanent magnets contain rare earth elements that are subject to supply chain risks, there has been increasing interest in discovering rare-earth-free permanent magnets [58]. We simplify the problem of finding such a magnet to finding materials with high magnetic density and a low Herfindahl–Hirschman index (HHI). According to the U.S. Department of Justice and the Federal Trade Commission, a material with an HHI score below 1500 is considered to have low supply chain risk [59]. Thus, we ask the model to generate materials with a magnetic density of 0.2 $Å^{-3}$ and an HHI score of 1250.

In Fig. 6(a), we observe that MatterGen generates S.U.N. structures that are narrowly distributed around the target values, despite the labeled fine-tuning data being extremely scarce in that region. Compared to a model that only targets high magnetic density values (single), targeting both properties (joint) shifts the distribution of HHI scores closer towards the desired target value while retaining high magnetic density values. Fig. 6(b) showcases the effect of jointly targeting both properties on the distribution of elements found in the generated materials. Due to the lower HHI scores, elements commonly employed for high-magnetic density materials that have supply chain issues, e.g., Cobalt (Co) and Gadolinium (Gd), are practically absent in the jointly generated structures. In contrast, these elements are still present in structures generated by the model only targeting materials with high magnetic density (single).

3 Discussion

Generative models are particularly promising for tackling inverse design tasks as they can potentially explore entirely *novel* structures with desired properties in an efficient way. However, generating the 3D structure of stable crystalline materials is challenging due to their periodicity and the interplay between atom types, coordinates, and lattice. MatterGen improves upon limitations of previous methods [4, 56] by introducing a joint diffusion process for atom types, coordinates, and lattice (Section 2.1 and Appendices A.5 to A.7), which—combined with the substantially larger training dataset Alex-MP-20—drastically increase the stability, uniqueness, and novelty of generated materials. Thanks to the introduction of the adapter modules (Appendix B.1), MatterGen can further be fine-tuned to generate S.U.N. structures satisfying target constraints across a wide range of properties, with performance improvements over widely-employed methods such as MLFF-assisted RSS and substitution (Section 2.3), as well as ML-assisted screening (Section 2.5).

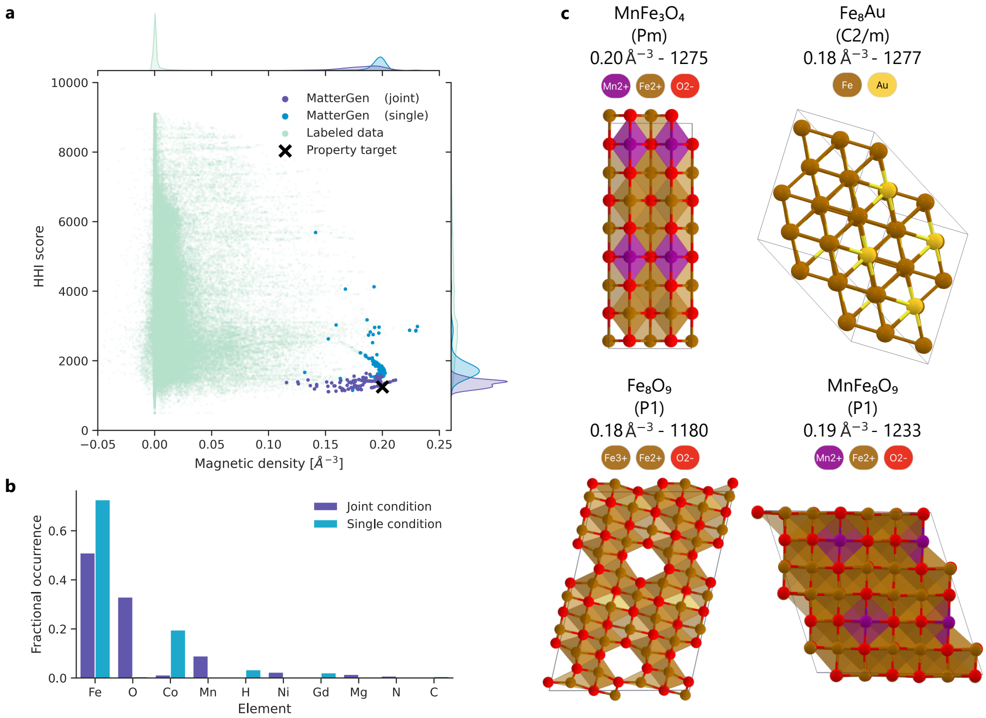

Fig. 6: **Designing low-supply-chain-risk magnets. (a)** Distribution of S.U.N. structures generated by MatterGen when fine-tuned on the HHI score (single) and on both HHI score and magnetic density (joint), as well as structures from the labeled fine-tuning dataset. The property target of MatterGen is denoted as a black cross. **(b)** Occurrence of most frequent elements in S.U.N. structures for the two fine-tuned MatterGen models. **(c)** S.U.N. structures on the Pareto front for the joint property optimization task, along with their chemical composition, space group, magnetic density, and HHI score.

Despite these advances, MatterGen could still be improved in several ways. For example, we observe that the model disproportionately generates structures with P1 symmetry compared to the training data, indicating a tendency for generating less symmetric structures, especially for larger crystals (see discussion in Appendix D.2). We hypothesize that further improvements on the denoising process, the backbone architecture, and the expansion of the training dataset could enable the model to overcome such issues. We also acknowledge that our extensive evaluations only cover some of the criteria required for real-world applications, with experimental validation and characterization being the ultimate test [57]. We discuss the challenges in evaluating the quality of crystalline materials from generative models in Appendix D.2.

Overall, we believe that the breadth of MatterGen's capabilities and the quality of generated materials represent a major advancement towards creating a universal generative model for materials with high real-world impact. Given the transformative effect of generative models in domains like image generation [60] and protein design [61], we envision that generative models like MatterGen will have a major impact in materials design in the coming years. As such, we are excited about the many directions in which MatterGen could be further extended. For instance, MatterGen could be expanded to cover a broader class of materials ranging from catalyst surfaces to metal organic frameworks, enabling us to tackle challenging problems like nitrogen fixation [62] and carbon capture [63]. The property constraints can be extended to non-scalar quantities like the band structure or X-ray diffraction (XRD) spectrum, which would further enable applications ranging from band engineering to the prediction of atomic structures of experimentally-measured XRD spectra of unknown samples.

Supplementary information Additional explanations and details regarding the MatterGen architecture, fine-tuning approach, datasets, and results can be found in the supplementary information.

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Declarations

Author contributions

AF, MH, RP, RT, TX, CZ and DZ (alphabetically ordered) conceived the study, implemented the methods, performed experiments, and wrote the manuscript. XF led the development of the adapter modules. SS implemented and ran the symmetry conditioned generation. JS implemented the band gap workflow. BN proposed the task of low-supply-chain risk magnets. ZL, YZ, HY, HH, and JL developed the machine learning force field. XF, SS, JC, LS, JS, BN, HS, SL, C-WH, ZL, YZ, HY, HH, and JL helped with implementing the methods, conducting experiments, and writing the manuscript. TX and RT led the research.

Appendix

Table of Contents

A Diffusion model for periodic materials

This section contains additional model details, following the notation listed in Table A1.

General notation	
$n \in \mathbb{N}$	Number of atoms in a crystal
$\boldsymbol{M} = (\boldsymbol{X}, \boldsymbol{A}, \boldsymbol{L})$	A crystal structure
$\boldsymbol{X} \in [0,1)^{3\times n}$	Fractional atomic coordinates
$\tilde{\boldsymbol{X}} \in \mathbb{R}^{3\times n}$	Cartesian atomic coordinates
$\boldsymbol{A} \in \mathbb{A}^n$	Atomic species in a crystal
$\boldsymbol{L} = (\boldsymbol{l}^1, \boldsymbol{l}^2, \boldsymbol{l}^3) \in \mathbb{R}^{3\times 3}$	The unit cell lattice matrix
$\boldsymbol{l}^j \in \mathbb{R}^3$, $j \in \{1,2,3\}$	The j-th lattice vector
$\boldsymbol{V} = (\boldsymbol{v}_1, \boldsymbol{v}_2, \dots \boldsymbol{v}_N) \in \mathbb{R}^{d\times N}$	Concatenation of N d-dimensional column vectors into a matrix
$\mathcal{E} \subset \{1,2,\dots,n\}^2 \times \mathbb{Z}^3$	Set of edges in a material
$i, j \in \{1,2,\dots,n\}$	Index of an atom in a material
$d \in \mathbb{N}$	The number of hidden dimension in our GNN
$\mathbf{1}_n \in \mathbb{R}^n$	n-dimensional column vector containing ones
Diffusion notation	
$t \in 1, 2, \dots, T$	Diffusion timestep
$T \in \mathbb{N}$	Number of time discretization steps for the diffusion process
$q(\boldsymbol{x}_0)$	The data distribution
$q(\boldsymbol{x}_t\|\boldsymbol{x}_{t-1})$	Single-step diffusion transition kernel
$q(\boldsymbol{x}_t\|\boldsymbol{x}_0)$	One-shot diffusion kernel
$q(\boldsymbol{x}_T)$	Prior (noise) distribution
$\boldsymbol{s}_{\boldsymbol{\theta}}(\cdot, t)$	Score model
$\boldsymbol{s}_{\boldsymbol{X},\boldsymbol{\theta}}(\cdot, t)$	Score model for atomic coordinates
$\log p_{\boldsymbol{\theta}}(\boldsymbol{A}_0\|\boldsymbol{X}_t, \boldsymbol{L}_t, \boldsymbol{A}_t, t)$	Predicted logits for atom types at $t = 0$.
$\boldsymbol{s}_{\boldsymbol{L},\boldsymbol{\theta}}(\cdot, t)$	Score model for lattice
$\boldsymbol{z}$	Standard Gaussian noise $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$

Table A1: Table of notations

A.1 Representation of periodic materials

Any crystal structure can be represented by some repeating unit (called the *unit cell*) that tiles the entire 3D space. The unit cell itself contains a number of atoms that are arranged inside of it. Thus, we use the following universal representation for a material $\boldsymbol{M}$:

$$\boldsymbol{M} = (\boldsymbol{A}, \boldsymbol{X}, \boldsymbol{L}), \tag{A1}$$

where $\boldsymbol{A} = (a^1, a^2, \dots, a^n)^\top \in \mathbb{A}^n$ are the atomic species of the atoms inside the unit cell; $\boldsymbol{L} = (\boldsymbol{l}^1, \boldsymbol{l}^2, \boldsymbol{l}^3) \in \mathbb{R}^{3\times 3}$ is the lattice, i.e., the shape of the repeating unit cell; and $\boldsymbol{X} = (\boldsymbol{x}^1, \boldsymbol{x}^2, \dots, \boldsymbol{x}^n) \in [0,1)^{3\times n}$ are the *fractional* coordinates of the atoms inside the unit cell.

The lattice $\boldsymbol{L}$ is a parallelepiped defined by the three lattice vectors $\boldsymbol{l}^1, \boldsymbol{l}^2$, and $\boldsymbol{l}^3$. It can thus be compactly represented as a single 3×3 matrix with the three lattice vectors as its columns. The volume of a lattice is given by $\mathrm{Vol}(\boldsymbol{L}) = |\det \boldsymbol{L}|$. Any